Mansoura University
Faculty of Computer & Information Sciences
Information Technology Department

Course: Selected-1 IT321P
Level:3rd Information Technology-programs
Question Bank

Digital Image Processing

Question Bank

Part 1

1. **………… is the process of manipulating an image so that the result is more suitable than the original for a specific application.**
   a) image enhancement — c) Image Restoration
   b) Image Sharpening — d) Image Blurring
2. **Image ………… is a subjective process, while image ………… is an objective process.**
   a) Enhancement / Restoration — c) Restoration / Enhancement
   b) Sharpening / Smoothing — d) Smoothing / Sharpening
3. **Recognition is the process that assigns a ………… to an object based on its descriptors.**
   a) Label — c) Weight
   b) Tag — d) Value
4. **The quality of the digital image is determined by …………**
   a) The number of samples — c) The number of intensity levels
   b) All of these — d) None of these
5. **A basic tool used in tasks such as zooming, shrinking, and rotating is called …………**
   a) Segmentation — c) Filtering
   b) Morphological processing — d) Interpolation
6. **Consider 64x64 intensity image, the number of bits used in this image equals ………… bits.**
   a) 32,768 — c) 98,304
   b) 262,144 — d) 4096
7. **Two pixels p and q with values from V are ………… if q is in the set of ND(p) and the set of N4(p)∩N4(q) does not belong to V.**
   a) 4-adjacent — c) 8-adjacent
   b) m-adjacent — d) D-adjacent
8. **With ………… domain enhancement, techniques are based on direct manipulation of pixels in an image.**
   a) Frequency — c) Spatial
   b) All of these — d) None of these
9. **With frequency domain enhancement, techniques are based on modifying ………… of an image.**
   a) Neighbors — c) Bits
   b) The Fourier Transform — d) Pixels
10. **………… transform expand the values of high pixels while compressing the darker level values.**
    a) Log — c) Power law
    b) Inverse log — d) Slicing

11. **Many image capturing and printing devices uses ………… process to enhance the given image by power-law response phenomena.**
    a) Gamma correction   c) Gamma Thresholding
    b) Gamma correction   d) None of these
12. **The techniques used to increase the dynamic range of the intensity levels in the image is called …………**
    a) Thresholding   c) Gamma correction
    b) Slicing   d) Contrast stretch
13. **We could highlight the contribution made by each bit to the total image representation using …………**
    a) Gray-level slicing   c) Bit-plane slicing
    b) All of these   d) None of these
14. **The technique used to ………… a specific range of intensity levels in a given image is called gray-level slicing.**
    a) Smooth   c) Sharpen
    b) Filter   d) Highlight
15. **The method used to increase the dynamic range of the gray-level in a narrow-contrast image to cover full-range of gray level is called …………**
    a) Histogram equalization   c) Thresholding
    b) Quantization   d) Smoothing
16. **………… deals with techniques for reducing the storage required to save an image, or the bandwidth required to transmit it.**
    a) Compression   c) Overlapping
    b) Morphological processing   d) Segmentation
17. **………… techniques tend to be based on mathematical or probabilistic models of image Degradation.**
    a) Image enhancement   c) Image Restoration
    b) Image Sharpening   d) Image Blurring
18. **………… deals with tools for extracting image components that are useful in the representation and description of shape.**
    a) Compression   c) Overlapping
    b) Morphological processing   d) Segmentation
19. **Image representation and description almost always follow the output of ………… stage.**
    a) Segmentation   c) Recognition
    b) Compression   d) restoration
20. **Image ………… is the first process at digital Image Processing.**
    a) Acquisition   c) Selection
    b) Filtering   d) Quantization
21. **In ………… image, components of histogram are concentrated on the low side of the gray scale.**
    a) Bright   c) Low-contrast
    b) Dark   d) High-contrast
22. **………… transform expand the values of dark pixels while compressing higher values of lighter pixels.**
    a) Inverse Log   c) Power law
    b) Slicing   d) Log

23. In ………… image, histogram is narrow and centered towards the middle of the gray level.

a) Bright
b) Dark
c) Low-contrast
d) High-contrast

24. With ………… domain enhancement, techniques are based on modifying the Fourier transform of an image.

a) Spatial
b) None of these
c) Frequency
d) All of these

25. The ………… of the region R, is the set of pixels in the region that have one or more neighborhood that are not in R.

a) Path
b) Neighbors
c) boundary
d) None of these

Part 2

1. **In neighborhood operations working is being done with the value of image pixel in the neighborhood and the corresponding value of a subimage that has same dimension as neighborhood. The subimage is referred as …………**
   a) Filter　　　c) Mask
   b) Template　　　d) All of these
2. **The response for linear spatial filtering is given by the relationship …………**
   a) Sum of filter coefficient's product and corresponding image pixel under filter mask
   b) Difference of filter coefficient's product and corresponding image pixel under filter mask
   c) Product of filter coefficient's product and corresponding image pixel under filter mask
   d) None of these
3. **Smoothing filter is used for which of the following work(s)?**
   a) Blurring　　　c) Noise reduction
   b) All of these　　　d) None of these
4. **Which of the following filter(s) has the response in which the central pixel value is replaced by value defined by ranking the pixel in the image encompassed by filter?**
   a) Average filter　　　c) Laplacian filter
   b) Median filter　　　d) None of these
5. **An image contains noise having appearance as black and white dots superimposed on the image. Which of the following noise(s) has the same appearance?**
   a) Salt-and-pepper noise　　　c) Gaussian noise
   b) All of these　　　d) None of these
6. **While performing the max filtering, suppose a 3*3 neighborhood has value (10, 20, 20, 20, 15, 20, 20, 25, 100), then what is the value to be given to the pixel under filter?**
   a) 15　　　c) 20
   b) 100　　　d) 25
7. **A spatial domain filter of the corresponding filter in frequency domain can be obtained by applying which of the following operation(s) on filter in frequency domain?**
   a) Fourier transform　　　c) Inverse Fourier transform
   b) None of these　　　d) All of these
8. **Which of the following lowpass filters is/are covers the range of very sharp filter function?**
   a) Ideal lowpass filters　　　c) Butterworth lowpass filter
   b) Gaussian lowpass filter　　　d) All of these
9. **Butterworth lowpass filter has a parameter, filter order, determining its functionality as very sharp or very smooth filter function or an intermediate filter function. If the parameter value is very high, the filter approaches to which of the following filter(s)?**
   a) Ideal lowpass filter　　　c) Gaussian lowpass filter
   b) All of the mentioned　　　d) None of the mentioned
10. **In frequency domain terminology, which of the following is defined as "obtaining a high-pass filtered image by subtracting from the given image a lowpass filtered version of itself"?**
    a) Emphasis filtering　　　c) Unsharp masking
    b) Butterworth filtering　　　d) None of these
11. **Which of the following is/ are a generalized form of unsharp masking?**

a) Lowpass filtering
b) Emphasis filtering
c) High-boost filtering
d) All of these

12. In general, which of the following assures of no ringing in the output?
a) Gaussian Lowpass Filter
b) Butterworth Lowpass Filter
c) Ideal Lowpass Filter
d) All of the mentioned

13. The Laplacian is which of the following operator?
a) Nonlinear operator
b) Linear operator
c) Order-Statistic operator
d) None of these

14. What is the sum of the coefficient of the mask defined using gradient?
a) 1
b) 0
c) -1
d) None of these

15. While performing the median filtering, suppose a 3*3 neighborhood has value (10, 20, 20, 20, 15, 20, 20, 25, 100), then what is the value to be given to the pixel under filter?
c) 15
d) 100
c) 20
d) 25

16. What is the output of a smoothing, linear spatial filter?
a) Median of pixels
b) Minimum of pixels
c) Maximum of pixels
d) Average of pixels

17. Which of the following is the disadvantage of using smoothing filter?
a) Blur edges
b) Remove sharp transitions
c) Blur inner pixels
d) Sharp edges

18. Median filter belongs to which category of filters?
a) Linear spatial filter
b) Order static filter
c) Frequency domain filter
d) Sharpening filter

19. Which of the following is the primary objective of sharpening of an image?
a) Blurring the image
b) Increase the brightness of the image
c) Highlight fine details in the image
d) Decrease the brightness of the image

20. In spatial domain, which of the following operation is done on the pixels in sharpening the image?
a) Integration
b) Median
c) Average
d) Differentiation

21. The feature(s) of a highpass filtered image is/are …………
a) Have less gray-level variation in smooth areas
b) Emphasized transitional gray-level details
c) An overall sharper image
d) All of these

22. A frequency domain filter of the corresponding filter in spatial domain can be obtained by applying which of the following operation(s) on filter in spatial domain?
a) Fourier transform
b) None of the mentioned
c) Inverse Fourier transform
d) All of the mentioned

23. Which of the following filtering is done in frequency domain in correspondence to lowpass filtering in spatial domain?
a) Gaussian filtering
b) High-boost filtering
c) Unsharp mask filtering
d) None of the mentioned

24. In a filter, all the frequencies inside a circle of radius D0 are not attenuated while all frequencies outside circle are completely attenuated. The D0 is the specified nonnegative distance from origin of the Fourier transform. Which of the following filter(s) characterizes the same?

a) Ideal filter
b) Gaussian filter
c) Butterworth filter
d) All of the mentioned

25. What is accepting or rejecting certain frequency components called as?

a) Filtering
b) Slicing
c) Eliminating
d) None of the Mentioned

Part 3

1. **Principle sources of noise arise during image.**
   a) Destruction — c) degradation
   b) Restoration — d) acquisition
2. **The purpose of restoration is to gain …………**
   a) Degraded image — c) The original image
   b) Pixels — d) Coordinates
3. **Degraded image is produced using degradation function and …………**
   a) Additive noise — c) Convolution
   b) All of these — d) None of these
4. **Filter that performs opposite to band rejected filter is called …………**
   a) Low-pass filter — c) bandpass filter
   b) high-pass filter — d) max filter
5. **Contra-harmonic mean filter produces …………**
   a) degraded image — c) original image
   b) restored image — d) plane
6. **Band-reject filters are used where the noise components are usually …………**
   a) Rejected — c) unknown
   b) Known — d) taken
7. **Frequencies in predefined neighborhood are rejected by …………**
   a) Notch filter — c) bandpass filter
   b) High-pass filter — d) Max filter
8. Filter that computes average between min and max value is called …………
   a) max filter — c) midpoint filter
   b) median filter — d) alpha-trimmed mean filter
9. **Image compression comprised of …………**
   a) Encoder — c) Decoder
   b) All of these — d) None of these
10. **If pixels can be reconstructed without error, mapping is called …………**
    a) Reversible — c) Irreversible
    b) Temporal — d) None of these
11. **Replication of pixels is called …………**
    a) spatial redundancy — c) temporal redundancy
    b) All of these — d) None of these
12. **step edge detection is between pixels over the distance of …………**
    a) 1-pixel — c) 2-pixel
    b) 3-Pixel — d) 4-pixel
13. **Second-derivatives approximation says that it is nonzero only at …………**
    a) Ramp — c) Step
    b) Onset — d) edges
14. **Discontinuity approach of image segmentation depends upon …………**
    a) low frequencies — c) smooth changes
    b) abrupt changes — d) contrast
15. **Information per source is called …………**
    a) Entropy — c) sampling

b) Quantization d) normalization

16. For line detection, we use mask that is …………

a) Gaussian c) Laplacian
b) Ideal d) Butterworth

17. Second derivatives in image segmentation produce …………

a) Thick edges c) thin edges
b) Fine edges d) no edges

18. Gradient of image is obtained through …………

a) Differentiation c) summation
b) Subtraction d) integration

19. Examples of region approach in segmentation are in …………

a) Region growing c) Region splitting
b) All of these d) None of these

20. Compressed image can be recovered back by …………

a) Image enhancement c) image decompression
b) Image contrast d) image equalization

21. Digital video is a sequence of …………

a) Frames c) Matrix
b) Pixels d) Coordinates

22. Information is …………

a) Data c) Raw data
b) Meaningful data d) All of these

23. Every run length code introduces new …………

a) Frames c) Pixels
b) Matrix d) Intensities

24. In the coding redundancy technique, we use

a) Fixed length code c) Variable length code
b) All of these d) None of these

25. Information ignored by the human eyes represents …………

a) Coding redundancy c) Spatial redundancy
b) Temporal redundancy d) Irrelevant information